Business Requirements Document

Velora — Style, Decoded by Data

1. Purpose

This document outlines the business requirements for a project analyzing consumer shopping behavior for Velora, a retail fashion brand. The goal is to understand what drives customer purchase decisions and repeat buying — such as subscription status, review ratings, product category, age group, and shipping preferences — and to turn that into insights that support better marketing and sales decisions.

2. Business Problem

Velora wants to understand shifts in customer purchasing patterns across demographics, product categories, and customer subscription status. This project answers the question:

> *"How can Velora leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

3. Objectives

- Clean and prepare the raw consumer data for analysis.
- Identify key factors driving purchases and repeat buying (subscription status, review rating, product category, age group).
- Segment customers by age group and gender, and measure loyalty via subscription status.
- Compare performance across product categories (Clothing, Accessories, Footwear, Outerwear) and shipping types.
- Present findings through an interactive dashboard and a clear report/presentation.

4. Scope & Deliverables

Deliverable	Description
Data Preparation (Python)	Clean and transform the raw dataset — handle missing values, fix formatting, create useful new fields (e.g. age group, subscription flag).
Data Analysis (SQL)	Organize the cleaned data into tables, simulate transactions, and write queries to explore customer segments, subscription status, and purchase drivers.
Dashboard (Power BI)	Build an interactive dashboard (Velora Customer Behaviour Dashboard) showing total customers, average purchase amount, review ratings, revenue/sales by category, and purchase behavior by age group.
Report & Presentation	A short written report of key findings and recommendations, plus a slide deck to present insights to stakeholders.

5. Key Questions This Project Should Answer

- Which age group buys the most, and how does subscription status relate to overall spend and revenue?
- What share of customers are subscribed, and how does that relate to average spend and total revenue?
- Which products achieve the highest average review ratings?
- Which product category generates the most revenue and sales volume, and which products sell the most within each category?
- Which age group spends the most, and how does average purchase amount compare between shipping types (e.g. Standard vs. Express)?
- How does total revenue compare between male and female customers?
- Are repeat buyers (more than 5 previous purchases) more likely to be subscribed?
- How are customers distributed across New, Returning, and Loyal segments based on purchase history?
- Which products depend most on discounts, and which discount users still spend above the average purchase amount?

6. Tools & Data

Tools: Python (data cleaning), SQL (data modeling & queries), Power BI (dashboard), Word/PowerPoint (report & presentation).

Data: A consumer shopping behavior dataset including customer demographics (age, gender), subscription status, product category, shipping type, purchase amount, and review rating.

7. Assumptions & Constraints

- The dataset provided is a representative, static (historical) snapshot — not a live data feed.
- “Simulated business transactions” means structuring the data into a transactional table format for SQL analysis, not connecting to a live system.
- Subscription status (Yes/No) is used as the main proxy for customer loyalty.
- Any personal/identifying data is anonymized before analysis.

8. Success Criteria

- Cleaned dataset with no major data quality issues, ready for SQL loading.
- SQL queries that clearly answer the key questions in Section 5.
- A working Power BI dashboard covering segments, channels, and purchase drivers.
- A short report and presentation with at least 3 clear, actionable recommendations.